

Muhibu KANEKA

Geo-environmental and spatial data Analyst

PROFILE

Geo-environmental Spatial data analyst with a background in geology and a strong interest in the mining and natural-resources sector. Experienced in geospatial analysis, spatial data management, GIS, remote sensing and workflow automation, supporting multidisciplinary environmental related projects across office and field settings.

CONTACT

13100 Aix-en-Provence, France

+33 783 513 354

muhibu.kaneka@gmail.com

LinkedIn

Webpage

TECHNICAL SKILLS

- ArcGIS Pro, QGIS, ENVI, Geo, Datamine
- PostgreSQL, PostGIS, SQL
- Geoserver, OpenLayers, Leaflet
- Python, HTML, CSS, JS
- Git (Github/Gitlab)
- PowerBI

CORE COMPETENCIES

- Spatial Database Development & Management
- Spatial Analysis (Qualitative & Quantitative)
- Remote Sensing & Satellite Image Processing
- Field Data Collection & Ground Truthing
- Data Quality Control & Validation
- Web mapping & Data Visualization

SOFT SKILLS

- Aadaptability
- Detail oriented
- Quick learner
- Strong communication skills

LANGUAGES

- Kiswahili : Native
- English : Fluent
- French : Fluent

HOBBIES

- Cycling
- Meditation
- Theatre
- Safari

Rreferee

Oswald Bomani

Hydrogeologist, Barrick

+255 757 116 317

Oswald.Bomani@barrick.com

EXPERIENCES

Geospatial analyst

INRAE, Aix-en-Provence | oct 2021 - current

Context :

- Development of an interactive web mapping application.

Key Responsibilities:

- Designed and administered spatial and non-spatial databases (PostgreSQL/PostGIS), ensuring data integrity and consistency.
- Extracted, cleaned, transformed, and loaded data from multiple sources (IGN, DFCI, CRIGE, BDIFF) into internal spatial databases.
- Built and maintained a Web GIS application to visualize spatial data and support stakeholder decision-making.
- Collaborated closely with multidisciplinary teams, defined data needs, delivering GIS outputs, and support project objectives.
- Prepared technical reports, GIS documentation, and user manuals for project teams and partners.

Geospatial analyst

INRAE, Aix-en-Provence | Mar - Sept 2021

Context :

- The development of an automated satellite image processing system for forest fire mapping and monitoring

Key Responsibilities:

- Improved a python-based workflows which automate satellite image processing, spatial analysis, and fire-mapping outputs.
- Developed customized workflows and automated processes to improve efficiency in data processing and visualization.
- Performed manual analysis of satellite images using GIS and remote sensing techniques, mirroring the automated workflow to test and validate its accuracy.
- Created an automation script for raster-to-vector format conversion and integrated the results into a spatial database.
- Conducted ground truthing for the verification of the efficiency of the automation tool

Remote sensing analyst

ESPACE (UMR 7300), Nice | Mar - July 2020

Context:

- The analysis of runoff due to suburban development on the “Brague catchment” (southern France)

Key Responsibilities:

- Processed satellite imagery to generate land cover and land-use maps
- Conducted spatial and statistical analyses to assess the impact of suburban development on runoff dynamics.
- Developed hydrological models incorporating runoff coefficients, precipitation, and infiltration parameters.
- Produced maps, analytical reports, and visual outputs to support environmental risk assessment.

Geologist (internship)

TPDC, Tanzania | Jan - Apr 2019

Context:

- Gravity prospecting for structural studies using remote sensing data in the southern Tanzania petroleum basins

Key Responsibilities:

- Conducted satellite gravity data acquisition, processing, and interpretation for structural analysis.
- Performed structural mapping (faults, folds, lineaments) using GIS and remote sensing techniques.
- Produced maps of structural mapping of faults, folds, and lineament identification and analysis.
- Produced maps of anomalous behavior, including Bouguer gravity anomalies.

EDUCATION

Certificate of Data analysis	M2i, France	Jun - Sept, 2025
MSc. Environmental hazards and risks management	Côte d'Azur university, France	2019 - 2021
MSc. Georesources	Tlemcen university, Algeria	2016 - 2018
BSc. Geoscience	Tlemcen university, Algeria	2012 - 2015

CERTIFICATES & AWARDS

- *Geospatial applications for risk management | UNOOSA and CSSTEAP*
- *Leadership and Conflict management and resolution | Excellencia training center*
- *Python for spatial analysis | Coursera*
- *Project management | LAMAABE*
- *French government scholarship |2019*
- *Algerian government scholarship | 2011*



Muhibu KANEKA

Geo-environment specialist
and spatial data Analyst

PROFILE

Geo-environment specialist with expertise in geospatial analysis, database management, GIS, remote sensing, and spatial visualization, experienced in building spatial databases, automating workflows, and supporting multidisciplinary environmental and risk-related projects in both office and field settings.

CONTACT

13100 Aix-en-Provence, France
+33 783 513 354
muhibu.kaneka@gmail.com
LinkedIn

TECHNICAL SKILLS

- ArcGIS Pro, QGIS, ENVI, Geo, Datamine
- PostgreSQL, PostGIS, SQL
- Geoserver, OpenLayers, Leaflet
- Python, HTML, CSS, JS
- Git (Github/Gitlab)
- PowerBI

CORE COMPETENCIES

- Spatial Database Development & Management
- Spatial Analysis (Qualitative & Quantitative)
- Remote Sensing & Satellite Image Processing
- Field Data Collection & Ground Truthing
- Data Quality Control & Validation
- Web mapping & Data Visualization

SOFT SKILLS

- Aadaptability
- Detail oriented
- Quick learner
- Strong communication skills

LANGUAGES

- Kiswahili : Native
- English : Fluent
- French : Fluent

HOBBIES

- Cycling
- Meditation
- Theatre
- Safari

Rreferee

Oswald Bomani
Hydrogeologist, Barrick
+255 757 116 317
Oswald.Bomani@barrick.com

EXPERIENCES

Geospatial analyst
INRAE, Aix-en-Provence | 2021 - 2025

Context :
- Development of an interactive web mapping application.

Key Responsibilities:
- Designed and administered spatial and non-spatial databases (PostgreSQL/PostGIS), ensuring data integrity and consistency.
- Extracted, cleaned, transformed, and loaded data from multiple sources (IGN, DFCI, CRIGE, BDIFF) into internal spatial databases.
- Built and maintained a Web GIS application using Django to visualize spatial data and support stakeholder decision-making.
- Collaborated closely with multidisciplinary teams to define data needs, deliver GIS outputs, and support project objectives.
- Prepared technical reports, GIS documentation, and user manuals for project teams and partners.

Geospatial analyst
INRAE, Aix-en-Provence | Mar - Sept 2021

Context :
- The development of an automated satellite image processing system for forest fire mapping.

Key Responsibilities:
- Improved a python-based workflows which automate satellite image processing, spatial analysis, and fire-mapping outputs.
- Developed customized workflows and automated processes to improve efficiency in data processing and visualization.
- Performed manual analysis of satellite imagery using GIS techniques, mirroring the automated workflow to test and validate its accuracy.
- Created an automated script for raster-to-vector format conversion and integrated the results into a spatial database (WKT format).
- Conducted ground truthing for the verification of the efficiency of the automation tool

Remote sensing analyst
ESPACE (UMR 7300), Nice | Mar - July 2020

Context:
- The analysis of runoff due to suburban development on the “Brague catchment” (southern France)

Key Responsibilities:
- Processed satellite imagery to generate land cover and land-use maps
- Conducted spatial and statistical analyses to assess the impact of suburban development on runoff dynamics.
- Developed hydrological models incorporating runoff coefficients, precipitation, and infiltration parameters.
- Produced maps, analytical reports, and visual outputs to support environmental risk assessment.

Geologist (internship)
TPDC, Tanzania | Jan - Apr 2019

Context:
- Gravity prospecting for structural studies using remote sensing data in the southern Tanzania petroleum basins

Key Responsibilities:
- Conducted satellite gravity data acquisition, processing, and interpretation for structural analysis.
- Performed structural mapping (faults, folds, lineaments) using GIS and remote sensing techniques.
- Produced maps of structural mapping of faults, folds, and lineament identification and analysis.
- Produced maps of anomalous behavior, including Bouguer gravity anomalies.

EDUCATION

Certificate of Data analysis	M2i, France	Jun - Sept, 2025
MSc. Environmental hazards and risks management	Côte d'Azur university, France	2019-2021
MSc. Georesources	Tlemcen university, Algeria	2016 - 2018
BSc. Geoscience	Tlemcen university, Algeria	2012 - 2015
Advanced certificate level	Kibaha Secondary school , Tanzania	2009 - 2011
Ordinary level certificate	Mzumbe Secondary school, Tanzania	2005 - 2008

CERTIFICATES & AWARDS

- *Geospatial applications for risk management | UNOOSA and CSSTEAP*
- *Leadership and Conflict management and resolution | Excellencia training center*
- *Python for spatial analysis | Coursera*
- *Project management | LAMAABE*
- *French government scholarship |2019*
- *Algerian government scholarship | 2011*



Muhibu KANEKA

I CONTACT

2 rue Jean-pierre Petit, 31700, Blagnac.

+255 747167400 / +33 783513354

muhibu.kaneka@gmail.com

LinkedIn

I SOFTWARE

- Python, SQL
- Power BI,
- QGIS, ArcGIS
- PostgreSQL/PostGIS
- Geoserver, OpenLayers, leaflet
- Git
- MS Word, MS Excel

I SOFT SKILLS

- Deligence
- Aadaptable
- Persistent
- Quick learner

I LANGUAGES

Kiswahili :	Native
English :	Fluent
French :	Fluent

I HOBBIES

- Cycling
- Meditation
- Theatre
- Safari

Geospatial data analyst

EXPERIENCES

Geospatial analyst and developer

INRAE, Aix-en-Provence | Oct 2021 - current

Context :

- Development of an interactive web mapping application and risks analysis decision tool.

Tasks performed:

- Collaborated with stakeholders including insurance companies, government agencies, and local communities to improve fire risk management strategies.
- Analyzed spatial data such as satellite imagery, topographic information, and historical fire records to assess fire vulnerability and impacts.
- Developed an interactive web mapping application for historical forest fire visualization to support decision-making processes.

Geospatial analyst

INRAE, Aix-en-Provence | Mar - Sept 2021

Context :

- The development of an automated satellite image processing system for forest fire mapping.

Tasks performed:

- Developed a Python script to automate fire mapping processes and analysis.
- Remote sensing analysis of satellite images using pixel-based and object-based techniques (OBIA).
- Developed a python script for vectorizing raster data of the fire burned area, and store the results into a database (WKT format).
- Conducted ground truthing for the verification of the automated data with field data

Remote sensing analyst

ESPACE (UMR 7300), Nice | Mar 2020- July 2020

Context:

- The analysis of runoff due to suburban development on the “Brague catchment” (southern France)

Tasks performed:

- Satellite imagery processing for creating land cover maps and produced the analysis
- Conducted a comprehensive study based on the influence of suburban development to urban runoff within the catchment area.
- Analyzed statistical data, and generating hydrological models encompassing runoff coefficients, precipitation, and infiltration.
- Presented a meaningful interpretation of the assessment and recommended potential solutions.

Assistant Geologist

Yapi Merkez, Dar-es-Salaam | July 2018- August 2019

Context:

- Mapping and assessment of natural hazards and vulnerability, with a focus on flooding and landslides.

Main duties performed :

- Processed and analyzed satellite imagery and statistical data to investigate vulnerabilities.
- Conducted vulnerability assessments and created natural hazard maps.
- Engaged in the mapping of natural hazards, database maintenance, and the preparation of comprehensive technical reports.

EDUCATION

MSc. Environmental hazards and risks management

Côte d'Azur university, France | 2019-2021

MSc. Geporessources

Tlemcen university, Algeria | 2012 - 2015

BSc. Geoscience

Tlemcen university, Algeria | 2012 - 2015

CERTIFICATES & AWARDS

- *Geospatial applications for risk management* | ***UNOOSA and CSSTEAP***
- *Leadership and Conflict management* | ***Excellencia training center***
- *Python for spatial analysis* | ***Coursera***
- *Project management* | ***LAMAABE***
- ***French government scholarship |2019***
- ***Algerian scholarship for African students | 2011***



Muhibu KANEKA

I CONTACT

20 rue de Cuques , 13100
Aix-en-Provence, France
0783 513 354
muhibu.kaneka@gmail.com
LinkedIn

I SOFTWARE

- Django Framework
- Python, SQL, HTML, CSS, JS
- QGIS, ArcGIS, GEE
- PostgreSQL/PostGIS
- Geoserver, OpenLayers, leaflet
- Git (Github/Gitlab)

I SOFT SKILLS

- Deligence
- Aadaptable
- Persistent
- Quick learner

I LANGUAGES

Kiswahili :	Native
English :	Fluent
French :	Fluent

I HOBBIES

- Cycling
- Meditation
- Theatre
- Safari

Geospatial Analyst and Developer

EXPERIENCES

Geospatial analyst and developer
INRAE, Aix-en-Provence | Oct 2021 - Mar 2025

Context :
- Development of an interactive web mapping application.

Main duties performed :
- Developed a web mapping application using Django Framework.
- Programmed and integrated geospatial functionalities.
- Designed and implemented a database while also administering the operational database.
- Drafted technical documents and prepared user manuals.
- Maintained and updated the database.

Geospatial analyst
INRAE, Aix-en-Provence | Mar - Sept 2021

Context :
- The development of an automated satellite image processing system for forest fire mapping.

Main duties performed :
- Developed a Python script to automate fire mapping processes and analysis.
- Manually analyzed satellite images using pixel-based and object-based techniques (OBIA).
- Developed a python script for vectorizing raster data of the fire burned area, and store the results into a database (in a WKT format).
- Conducted ground truthing for the verification of the automated data with field data

Remote sensing analyst
ESPACE (UMR 7300), Nice | Mar 2020- July 2020

Context:
- The analysis of runoff due to suburban development on the "Brague catchment" (southern France)

Main duties performed :
- Satellite imagery processing for creating land cover maps and produced the analysis
- Conducted a comprehensive study based on the influence of suburban development to urban runoff within the catchment area.
- Analyzed statistical data, and generating hydrological models encompassing runoff coefficients, precipitation, and infiltration.

Assistant Geologist
Yapi Merkez, Dar-es-Salaam | July 2018- August 2019

Context:
- Mapping and assessment of natural hazards and vulnerability, with a focus on flooding and landslides.

Main duties performed :
- Processed and analyzed satellite imagery and statistical data to investigate vulnerabilities.
- Conducted vulnerability assessments and created natural hazard maps.
- Engaged in the mapping of natural hazards, database maintenance, and the preparation of comprehensive technical reports.

EDUCATION

MSc. Environmental hazards and risks management
Côte d'Azur university, France | 2019-2021

BSc. Geoscience
Tlemcen university, Algeria | 2012 - 2015

CERTIFICATES & AWARDS

- Geospatial applications for risk management | **UNOOSA and CSSTEAP**
- Leadership and Conflict management | **Excellencia training center**
- Python for spatial analysis | **Coursera**
- Project management | **LAMAABE**
- **French government scholarship |2019**
- **Algerian scholarship for African students | 2011**